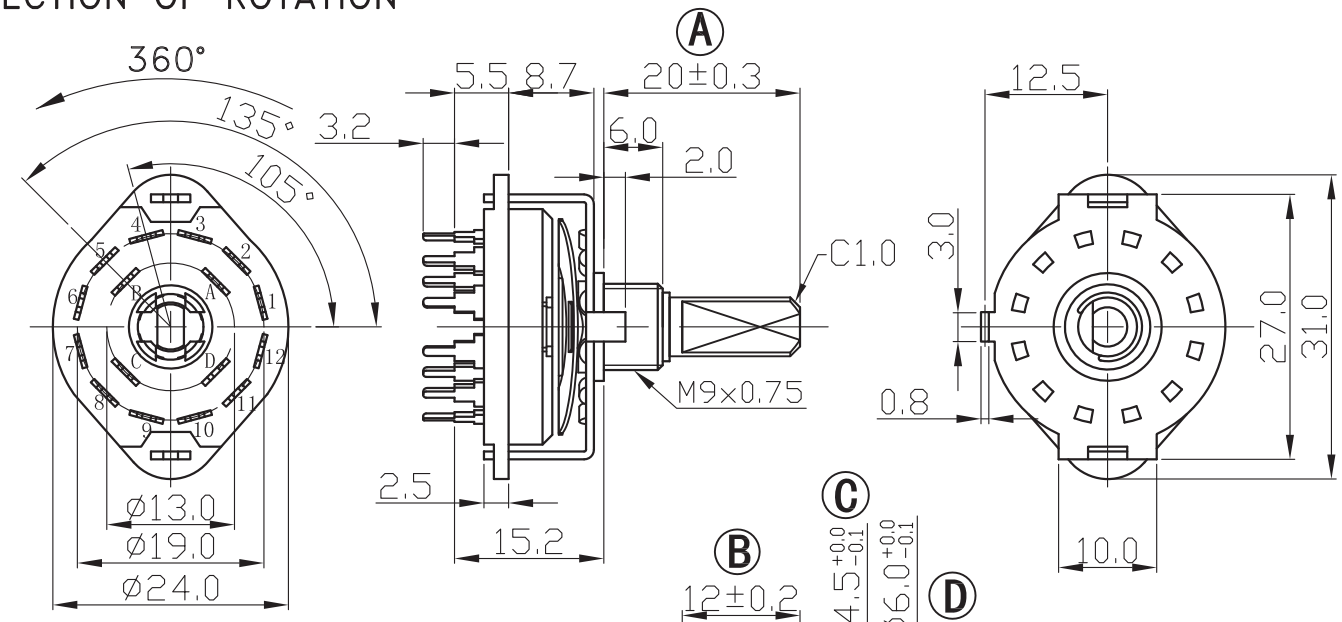
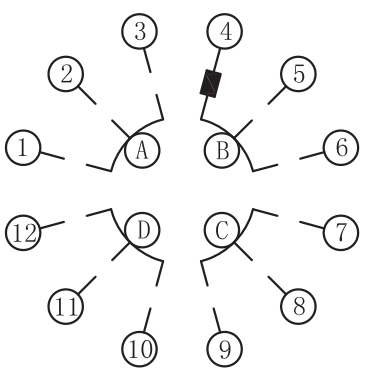
符合产品环境品质管理标准与用途

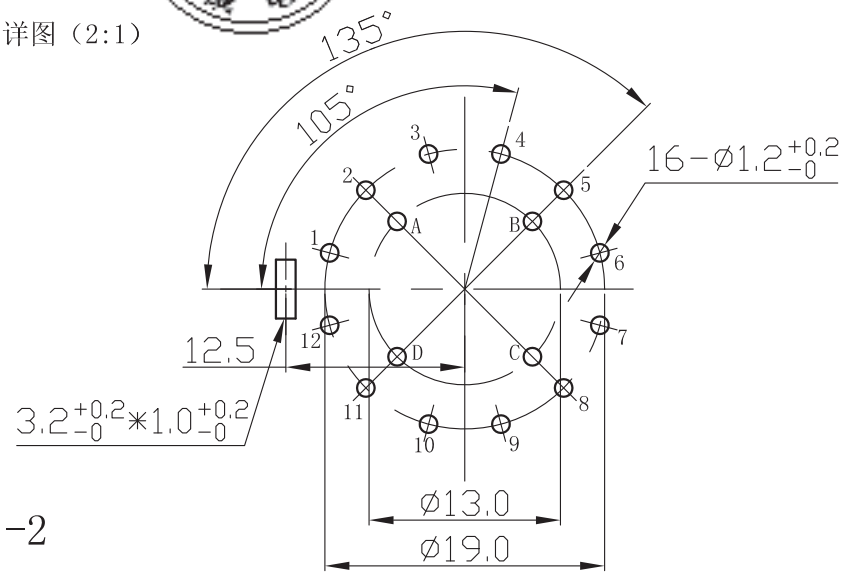
文件发行章

DIRECTION OF ROTATION





SWITCH CONTACT STATUS



P.C.B. MOUNTING HOLE DETAIL

角度规格:135°/105°-360°-2

接触方式:1-4-3

备注:1、Ⓐ-Ⓔ为重点管控尺寸。

01			03		
00	ORIGINAL DRAWING	2021-02-01	02		
. ISSU.	REVISION	DATE	. ISSU.	SOUNDWELL®	SOUNDWELL ELECTRONICS

			TOL. UNLESS OTHERWISE SPEC.		
蓝庆金	HUNG KAM PIU	HUNG KAM PIU	BASIC DIMENSIONS	TOL.	TITLE: SWITCH
			L≤10	±0.3	MODEL: RS240403A0X-HA1-020
DSGD.	CHKD.	APPD.	10<L	±0.5	
	SCALE		100≤L	±0.8	SPECIFICATION: FA200A120
	UNIT	mm	ANGLE	±5°	

RS24 SERIES SPECIFICATION
RS24系列规格书

1、 General 一般事项

1-1、Standard atmospheric conditions标准状态

Unless otherwise specified ,the standard range of atmospheric conditions for making measurements and test is as following limits:
除另有规定外，测量应在以下状态下进行：
Ambient temperature温度：15°C to 35°C
Relative humidity相对湿度：25% to 85%
Air pressure气压 ：86kPa to 106kPa

1-2、Operating temperature range
使用温度范围 ：-10℃ to 70℃

1-3、 Storage temperature range
保存温度范围 ：-40℃ to 85℃

2、Construction 构造

2-1、 Dimensions 尺寸
Refer to attached drawing 见所附成品图

3、 Rating 额定值

3-1、Rated voltage 额定电压: DC 10V

3-2、Maximum operating current (resistive load) 最大额定电流（阻抗负载）: 0.1A

4、 Electrical characteristics 电气性能

ITEM 项 目	CONDITIONS 条 件	SPECIFICATIONS 规 格
4-1、Contact resistance 接触电阻	Measured by the 1mA 5V D.C. voltage drop method 用DC 5V 1mA 电压降压法测定.	50mΩ Max 50mΩ 以下
4-2、Insulation resistance 绝缘电阻	Measurement shall be made under the condition which a voltage of 500V DC 1min is applied between individual terminals and bushing. 在端子和安装板间施加电压 500V DC 1分钟。	Between individual terminals and bushing 100MΩ MIN. 端子安装板间电阻100MΩ以上。
4-3、Dielectric strength 耐电压	A voltage of 500V AC shall be applied for 1 minute between individual terminals and bushing. 在端子和安装板间施加 AC 500V电压1分钟。	Without arcing or breakdown. 不得有绝缘破坏。
4-4、Timing 接触方式		☐ Shorting (短路型) ☑ No shorting (非短路型)
5、 Mechanical characteristics 机械性能		
5-1、Step angle 每段角度		30°±3°
5-2、Detent Torque 段点力矩		60± 40 mNm (600±400 gf.cm)
5-3、Push-pull strength of shaft 轴推拉强度	Push and pull static load of 8KG shall be applied to the shaft in the axial direction for 10S.(After soldering of the PC board) 在轴端,沿轴向施加8KG的静负荷力推和拉各10秒钟 (焊锡固定在PCB上）。	Without damage or excessive play in shaft No excessive abnormality in rotational feeling.And electrical characteristics shall be satisfied. 轴无破损,回转无异常.电气性能无异常.
5-4、Terminal strength 端子强度	A static load of 500gf shall be applied to the tip of terminals for 10S in any direction. 端子前端的任意方向施加500gf 的静负荷力10秒钟.	Without excessive play in terminal or poor contact. 端子不得有明显松动及接触不良。
5-5、Shaft wobble 轴摆动	A momentary load of 50mN.m(0.51Kgf.cm) shall be applied at the point 5mm from the tip of the shaft in a direction perpendicular to the axis of shaft. 在轴前端5mm处,沿径向瞬间施加50mN.m (0.51Kgf.cm)的力.	1.0x L / 30mm p-p MAX(L:shaft length) 1.0 x L / 30mm p-p 以下(L:安装长度)
5-6、Nut tightening strength 螺母紧固强度	Nut tighten with 10kgf.cm 螺母紧固强度为10kgf.cm.	No excessive uneven rotation feeling occur 不可有回转发生（打滑）
5-7、Side thrust strength of shaft 轴的垂直押引强度	A load of 20N(2.04Kgf) shall be applied at the point 5mm from the tip of the shaft in a direction perpendicular to the axis of shaft for 10S. 在轴前端5mm处加20N(2.04Kgf)的静负荷力10秒钟.	Without excessive play of bending in shaft No mechanical abnormally. 轴不得有明显松动及接触不良.

ITEM 项 目	CONDITIONS 条 件	SPECIFICATIONS 规 格
5-8、Shaft play in rotational wobble 轴的回转方向摆动	Testing by angle board. 用角度板测定.	4° MAX 4° 以下
5-9、Shaft stop strength 止档强度	The following torsion moment load of 5Kgf.cm shall be applied to the shaft for 5sec at both ends (after fixation) 固定后于轴心转动末端加5Kgf.cm力矩并持续5秒.	Electrical characteristics shall be satisfied with specification 电气性能符合规定要求
6、 Endurance characteristics 耐久性能		
6-1、Rotational life 回转寿命	The shaft of switch shall be rotated to 10,000 cycles at a speed of 600~800 cycles/H without electrical load,after which measurements shall be made. 在无负荷条件下轴以600~800周/小时速度回转 10,000 周。	Contact resistance: 1Ω Max 接触阻抗：1Ω以下 Detent Torque:No less than 50% of the initial torque . 段点力矩：不低于初期规格值的50%。
6-2、Solder ability 焊锡性	The terminals shall be immersed into solder bath at 260°C±5°C for 3s±1s in the same manner as para. 端子在260℃±5℃温度的焊锡槽内浸锡3秒±1秒。	A new uniform coating of solder shall 75% minimum of the surface being immersed. 浸渍面须有75%以上焊锡附着
6-3、Resistance to Soldering heat 耐焊接热	Manual soldering手工焊接 Bit temperature of soldering iron： 350°C less than Application time of soldering iron：within 3s 温度350°C以下，时间3秒以内。 Dip soldering槽焊 Printed wiring board : copper clad laminate board with thickness of 1.6mm. 使用基板：t=1. 6mm的覆铜板。 Preheating : 1、Surface temperature of board: 100°C or less. 2、Preheating time : within 1 min. 预热：基板表面温度100℃以下,时间1分钟以内。 Soldering : Solder temperature : 260±5°C or less Immersion time :within 3S 焊接：温度260±5℃或以下，时间3秒以内。	Electrical characteristics shall be satisfied No mechanical abnormality. 不得有绝缘体的破损、变形、接触 无异常。
6-4、 Damp heat 耐湿性	The swithc shall be stored at temprature of 40±2℃ with relative humidity of 90% to95% for 96±4H in a thermostatic chamber.And the encoder shall be subjected to standard atmosphericconditions for 1.5h After which measurements shall bo made. 温度40±2℃,湿度90~95%的恒温恒湿槽中放置96±4小时后,在常温常湿中放置1.5小时后测试.	Specifications in clause all items is shall be satisfied. 所有项应满足初期规格
6-5、 Dry heat 耐热性	The switch shall be stored at a temperature of 80±3℃ for 240±4H in a thermostatic chamber.And then the encoder shall be subjected to standard atmospheric conditions tor 1.5H.After witch measurement shall be made.温度80±3℃的恒温箱中放置240±4小时,常温、常湿放置1.5小时后测试.	Specifications in clause all items is shall be satisfied. 所有项应满足初期规格
6-6、Cold proof 低温特性	The switch shall be stored at a temperature of -25±3℃ for 96±4H in a thermostatic chamber.And then the encoder shall be subjected to standard atmospheric conditions tor 1.5H.After witch measurement shall be made.温度-25±3℃的恒温箱中放置96±4小时,常温、常湿放置1.5小时后测试.	Specifications in clause all items is shall be satisfied. 所有项应满足初期规格
7 Application Notes 使用上的事项		
Avoid storing the products in a place at high humidity and in Corrosive gases.please use this product with 12 months limitation. Ifany yemainder left after packing is opened,please store it with proper moistureproofing,gasproofing etc. 避名储藏于高温、高湿及腐蚀的场所。产品购入后需在12个月内使用完。拆包装后未使用完的剩余产品需储藏于防潮防毒的环境下。		

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